

VIBHA BHAVIKATTI

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EDUCATION

San Jose State University, San Jose, California
Master of Science in Computer Science

August 2024 - June 2026 (Spring 2026)

American University of Sharjah, Sharjah, UAE
Bachelor of Science in Computer Science, Minor in Data Science

September 2020 - June 2024

Dean's List Honoree: Fall 2020, Spring 2021, Fall 2021

Chancellor's List Honoree: Fall 2021

Relevant Coursework: Data Structures and Algorithms, Operating Systems, Software Engineering, Distributed Systems, Machine Learning, Database Systems, Artificial Intelligence, Deep Learning, NLP, Cloud Computing, LLM, Linear Algebra, Cybersecurity, Computer Networks

SKILLS

- **Programming:** Python, C++, Java, SQL, TypeScript, JavaScript
- **Data Systems:** Hadoop, Spark, Hive
- **Machine Learning & AI:** PyTorch, TensorFlow, Prompt Engineering, Reinforcement Learning, Computer Vision, LLM Evaluation, Model Evaluation
- **Systems & Infrastructure:** Multithreading, Distributed Systems, Linux, ETL Pipelines
- **Backend:** REST APIs, FastAPI, Flask
- **Cloud & DevOps:** AWS, Docker, CI/CD, Kubernetes
- **Databases:** PostgreSQL, MongoDB, Neo4j, Cassandra
- **Tools:** Git, Selenium, Locust, AI Coding Tools (Copilot, Claude Code)

EXPERIENCE

Scale AI

Aug 2025 - Present

Technical Advisor Intern - GenAI

- Built **evaluation pipelines** to analyze LLM outputs, identifying failure patterns, safety risks, and prompt injection vulnerabilities, and performed root-cause analysis to improve reliability and consistency across edge cases.
- Delivered **10+ high-priority evaluation tasks** under tight timelines, consistently completing rapid-turnaround projects to support model validation and deployment readiness.
- **Led technical reviews** across a **team of 4 interns**, improving evaluation quality, and test coverage.
- Developed automated workflows using **Selenium** and **Locust** to simulate real-world usage and adversarial inputs, enabling large-scale evaluation of model behavior.

Google Developer Groups (GDG) – SJSU

August 2025 – Present

Technical Lead

- Organize and lead AI and cloud engineering workshops for the GDG SJSU community (50+ members) while mentoring developers on **scalable system design** and **Google Cloud** technologies.

American University of Sharjah

Sept 2022 – May 2024

Teaching Assistant

- Mentored **100+ students** in Programming I, Programming Languages and Digital Systems across 4 semesters; provided hands-on support in lab sessions and code reviews.

PROJECTS

AutoRedTeam: LLM Red Teaming System | Python, NVIDIA Nemotron-253B, Ollama, LLM-as-Judge

Apr 2026 - Present

- Built an autonomous red teaming system using **Nemotron-Ultra-253B** (via NVIDIA NIM) to generate adversarial prompts across jailbreak, prompt injection, data extraction, and bias attacks on a local **Llama 3.2** model.
- Implemented an **LLM-as-judge** pipeline to score attack outcomes with severity ratings, achieving **50% success rate** and identifying a critical vulnerability in a 30-case audit.
- Designed a modular, config-driven system supporting model swapping and OpenAI-compatible APIs for testing different LLM endpoints.

CodeForge – Daily Coding Challenge Platform | React, TypeScript, Firebase, Docker, AWS, CI/CD

April 2025

- Built a full-stack **React/TypeScript web application** with **RESTful API** and **Firebase-based** user management and session state, delivering daily coding challenges to **30+ users**.
- Developed unit + E2E tests (Vitest, Cypress) in CI/CD pipelines; enforced automated quality gates with GitHub Actions.
- Deployed using Docker containers with **AWS services** (EC2, S3) and Render.

Big Data Systems Exploration and Applied Analytics

January 2025 - May 2025

- Developed **ETL pipelines** and **RESTful APIs** in Java 11 to ingest and process multi-million record datasets across distributed systems.
- Built **indexing and retrieval workflows** using Hadoop, Spark, and Hive to optimize queries and evaluate performance trade-offs.
- Designed schemas in **Hive Metastore** and coordinated distributed storage (**HBase**, **ZooKeeper**) with data validation and access control for scalable, fault-tolerant access.

Traffic Sign Detection and Localization (Capstone Project) | YOLOv3/YOLOv8, GCP, React, Python

Oct 2023 - May 2024

- Collaborated with **iEngineering** to build a detection system for traffic signs (YOLOv3/YOLOv8), achieving mAP@0.5 of **96.1%**.
- Built a **GCP-based** platform to upload, de-duplicate, and index large annotated datasets with geo-tagged, searchable metadata.
- Optimized inference for **real-time retrieval with <1s latency** using GPU acceleration and load testing on cloud infrastructure.

COMPETITIONS

KIBO Robot Programming Competition - JAXA and NASA

April 2023 – October 2023

- Simulated and developed Astrobree's laser-based ammonia leak detection and autonomous QR code scanning; final solution was executed aboard the **International Space Station (ISS)**, achieving **1st** place in the UAE and **6th** globally.